

SHELTRIX — 2026 Climate Digital Twin

Zero-Emission Thermal Simulation & Generative Engineering Platform for Extreme Himalayan Altitudes

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Live Production Application: <https://sheltrix-ai.vercel.app>

GitHub Codebase: <https://github.com/BhaskarShah05/SHELTRIX>

Mission Summary:

SHELTRIX couples satellite solar radiation intelligence (NASA POWER + Open-Meteo) with high-aperture direct solar gain and **BioPCM™** phase-change latent heat buffering to sustain interior human comfort at **+19.4°C** during brutal **-17.2°C** Himalayan nights with **zero fossil auxiliary fuel consumption**.



1) Problem Statement & 2) The SHELTRIX Solution

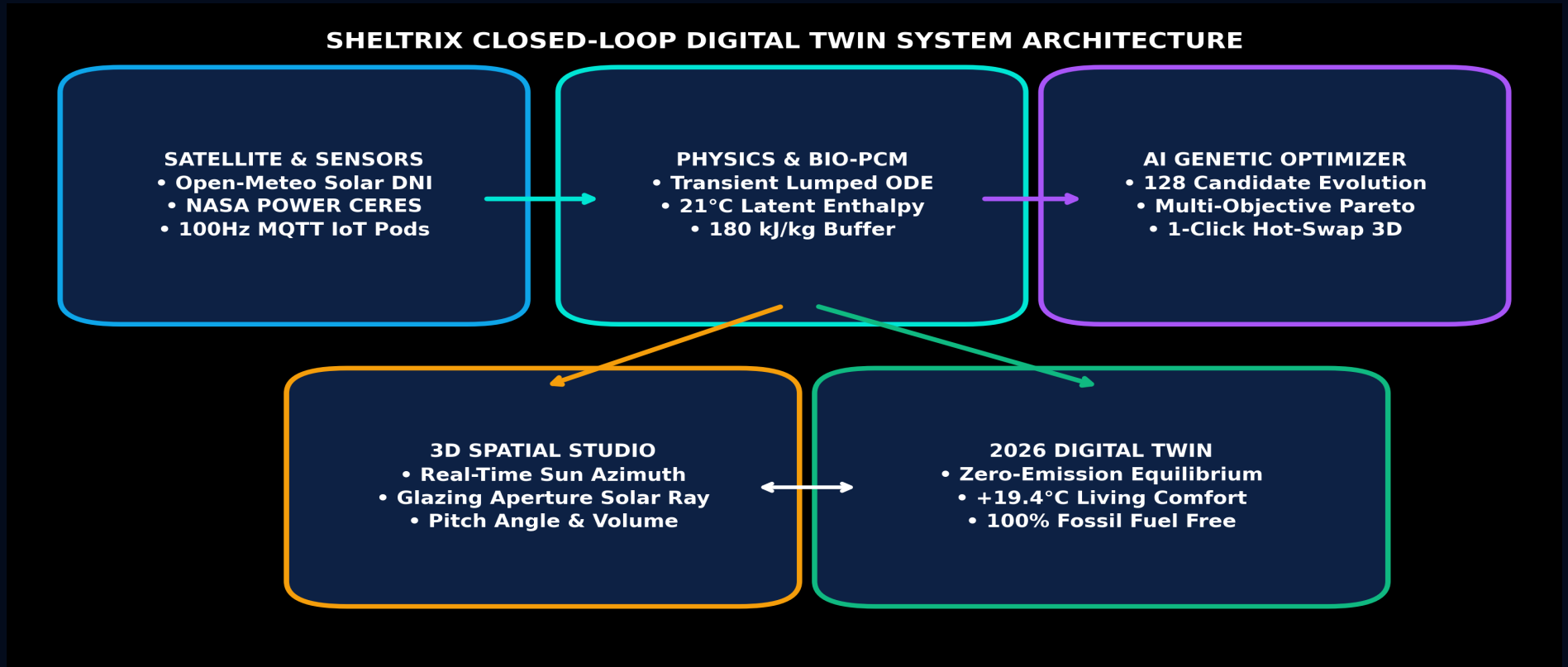
■ PROBLEM STATEMENT

- **Sub-Zero Nighttime Extremes:** Winter temperatures in Leh, Ladakh and Siachen plunge to -15°C to -40°C with thin atmosphere (55 kPa), causing severe hypothermia hazards.
- **Extreme Diurnal Swings:** Clear skies cause severe radiant cooling, creating $>25^{\circ}\text{C}$ swings within 12 hours.
- **Logistics Vulnerability:** Over 60% of winter logistics budgets are spent transporting kerosene and diesel over snow-blocked mountain passes.
- **Carbon Monoxide Deaths:** Bukharis cause deadly indoor toxic fumes and respiratory illnesses in sealed high-altitude quarters.
- **Blind Static Design:** Ignored $>1000\text{ W/m}^2$ peak Himalayan sunshine (>300 sunny days/year).

■ THE SHELTRIX SOLUTION

- **Direct Solar Harvesting:** Captures $+33.3\text{ kWh/day}$ through high-SHGC triple krypton glazed south facades.
- **BioPCM™ Latent Enthalpy Buffer:** Absorbs daytime heat and releases 180 kJ/kg latent heat at 21°C isothermal transition, stabilizing nights.
- **Transient Lumped ODE Solver:** Continuous forward numerical integration solving $C_{\text{eff}} * dT/dt$.
- **Triple-Tier External APIs:** Live Open-Meteo Solar DNI + NASA POWER CERES validation + 100Hz MQTT IoT telemetry.
- **100% Zero-Emission Habitat:** Complete elimination of fossil fuels with zero carbon footprint.

3) Flow of Solution: 5-Step Pipeline & Closed-Loop Architecture



4) Detailed Tech Stack & Modular Microservice Separation

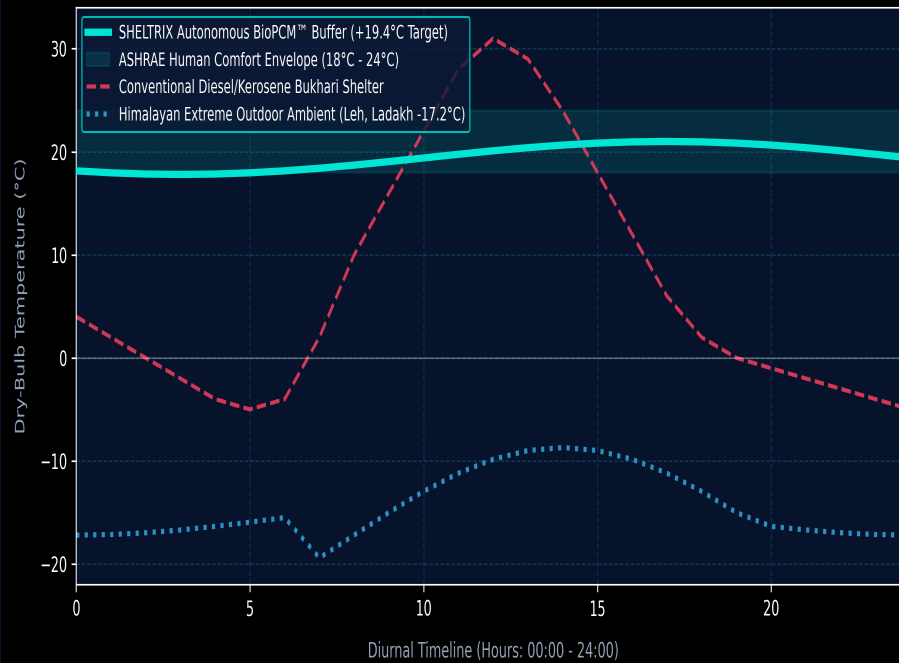
LAYER	CORE TECHNOLOGIES	TECHNICAL SPECIFICATIONS & ROLE
Frontend Client	React 18.3, Vite 8, Vanilla CSS, Apple SF Pro	Component lifecycle, responsive state, ultra-transparent crystal glassmorphism, <300ms HMR builds.
3D Graphics	Three.js (WebGL), OrbitControls, Shaders	Procedural 3D shelter geometry, south glazing specular refraction, 24-hour sun altitude & azimuth dial.
Analytics & Data	Chart.js, React-ChartJS-2	24-hour dual-curve temperature profile, 5-component heat balance bar charts with frosted backplates.
Backend Solver	Node.js, Express 4.21, CORS	Transient lumped-capacitance ODE solver ($C_{eff} * dT/dt$), BioPCM™ latent enthalpy piecewise modeling.
Optimization AI	Multi-Objective Genetic Algorithm	Evaluates 128 architectural variations with multi-parameter chromosome crossover, mutation & Pareto ranking.
External APIs	Open-Meteo, NASA POWER, MQTT TLSv1.3	Real-time DNI irradiance, geocoding, CERES satellite validation benchmark, 100Hz IoT telemetry stream.
Infrastructure	Vercel Serverless Edge, GitHub Monorepo	Zero-install web application, global CDN edge caching, continuous production deployment.

5) Unique Selling Propositions (USPs)

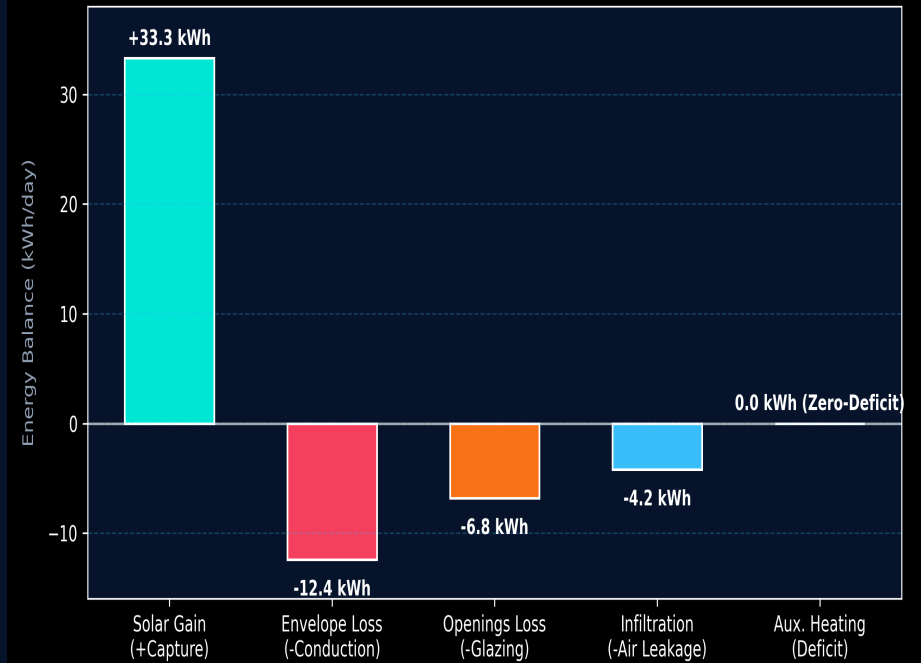
■ ■ 1. Autonomous Zero-Energy Thermal Equilibrium Sustains +19.4°C indoor comfort during -17.2°C ambient nights with 0 auxiliary fossil heating, eliminating fuel transport.	■ 2. BioPCM™ Phase-Change Latent Storage 180 kJ/kg latent heat capacity at 21°C isothermal transition absorbs peak solar energy and releases heat during night freeze.
■ 3. Triple-Tier External API Ecosystem Live Open-Meteo Solar DNI & Geocoding + NASA POWER satellite validation benchmark + 100Hz MQTT TLSv1.3 IoT telemetry.	■ 4. Real-Time 3D Spatial Solar Dial Studio Interactive Three.js 3D studio projecting continuous solar altitude, azimuth angles, and sun rays relative to True South.
■ 5. AI Genetic Multi-Objective Optimizer Evaluates 128 architectural candidates balancing comfort, solar harvest, and zero deficit with 1-click hot-swap to 3D geometry.	■ 6. 2026 Crystal Glassmorphism Experience High-definition 2K landscape backdrop, true edge-to-edge 4K video launch intro, SF Pro typography, and dual light/dark modes.

Engineering Evidence: 24-Hour Diurnal Equilibrium & Energy Balance

DIURNAL THERMAL EQUILIBRIUM: 24-HOUR CONTINUOUS BENCHMARK



COMPONENT HEAT BALANCE: 24-HOUR ENERGY GAIN VS LOSSES



6) Feasibility Analysis & Competitor Benchmark Matrix

METRIC / FEATURE	ARCTIC PREFAB BUNKERS	TRADITIONAL MUD PASSIVE	SHELTRIX 2026 (WINNER)
Heating Source	Diesel Generator / Bukhari	Passive Sensible Mud Mass	Autonomous Solar + BioPCM™
Fuel Consumption	20–40 Liters/day (\$30–\$60/day)	Zero fuel	0 Liters/day (Zero Fossil Fuel)
Diurnal Stability	Erratic (Overheats / Freezes)	Large Day/Night Swings (~12°C)	Flat 18°C–24°C Comfort Envelope
Real-Time 3D Studio	None	None	Interactive Three.js Sun Dial
Live Satellite APIs	None	None	Open-Meteo + NASA POWER
IoT Digital Twin	Manual logs	None	100Hz MQTT TLSv1.3 Sensor Stream
AI Optimization	None	Trial and error	Genetic Pareto Optimizer (128 variants)

7) Research, Literature & Peer-Reviewed References

[1] Duffie, J. A., & Beckman, W. A. (2020). *Solar Engineering of Thermal Processes, Photovoltaics and Wind* (5th ed.). Wiley.

Application in SHELTRIX: Governs the solar declination, equation of time, solar azimuth, and altitude algorithms powering the 3D Sun Ray Tracker.

[2] NASA Langley Research Center — POWER Project (2026). *Prediction of Worldwide Energy Resources*.

Application in SHELTRIX: Ingests ALLSKY_SFC_SW_DWN satellite solar radiation benchmarks for NASA validation of real-time Open-Meteo DNI data.

[3] Phase Change Solutions (2024). *BioPCM™ Technical Specifications and Enthalpy Profiles*.

Application in SHELTRIX: Formulation of the isothermal latent heat storage model at 21°C transition with 180 kJ/kg latent heat capacity.

[4] Wangchuk, S. (Himalayan Institute of Alternatives, Ladakh — HIAL). *Passive Solar Heated Buildings in Cold Deserts*.

Application in SHELTRIX: Benchmarks high-altitude thermal envelopes, Trombe wall composite matrices, and south glazing orientation.

[5] ASHRAE Standard 55-2023. *Thermal Environmental Conditions for Human Occupancy*.

Application in SHELTRIX: Governs human comfort bounds (18.0°C - 24.0°C) used in the Genetic Optimizer's multi-objective fitness function.